

PIR Sev 1 - no orders taken on UK and IE websites across all channels 25th March 2025

M&S Digital&Tech Post Incident Review



Incident/Problem Reference: INC000090685299

Incident Date: 25/03/25

PIR Date: 26/03/25

Author: Elaine Hawthorn

Technical Owner:  Sumitro Chakravarty

👉 *Analyses the root cause and provide technical recommendations to prevent incident recurrence.*

👉 *Develop and oversee the implementation of an action plan to address the root causes.*

👉 *Work closely with cross-functional teams to gather information and insights.*

PIR Reviewer:  Webster, Charly

👉 *Validates the root cause analysis, corrective actions, and preventive measures before signing off, ensuring alignment with business and operational standards.*

1. Incident Summary :

👉 *Provide a concise overview of the incident with brief update on issue, impact, root cause and resolution.*

Alerting identified that no orders were being taken on IE and UK across all channels. Customers would have got a 'sorry' message when trying to add to bag

Impact:

No orders taken between 05:00 and 05:29

Customers would have seen a 'sorry' page when trying to place an order

Spike in 5xx errors

Poor customer experience

As soon as the website recovered, double the expected volume of orders were placed indicating that there was no direct financial impact, the orders were displaced by 30 minutes

Various teams received alerts indicating that there was an increase in errors.

As soon as teams joined the bridge on the **Incident Channel**, it was identified that the problem was caused by a WCS release which was taking place.

During a WCS release, the blue/green deployment method is used which means that one of the environments (halls) is customer facing running the 'old' code whilst the other hall is being deployed to.

A pre-requisite for the deployment is to move customer traffic between halls using Akamai GTM (Akamai General Traffic Management). If the GTM is carried out correctly, there should be no customer traffic served when the pods are restarted. However, the GTM was wrongly configured which meant that customer traffic was still pointing at the hall which was about to be deployed to. Before the code deployment takes place, the pods are restarted and therefore a re-start occurred while the hall was still serving customer traffic. The re-start of the pods caused the drop in orders across all channels.

Once the root cause of the problem was identified, all of the customer traffic was moved via GTM to the hall which was still serving traffic. The traffic move completed at 05:29 and the incident was resolved. The WCS deployment was then able to proceed.

2. Incident Chronology :

👉 *Include a record of the data/time of key events. e.g. "07:36 Incident started, notified by ..." **keep fact based**".*

01:43 GTM initiated to move traffic to Hall 1

05:00 Pods on Hall 2 were restarted causing order dip

05:01 No orders placed

05:09 Post in Tech Ops saying that no orders placed

05:20 GTM initiated to move all traffic back to hall 1

05:24 Incident bridge opened

05:29 After successful GTM, confirmed that orders were getting placed

06:02 Confirmed that double the amount of orders placed in period after restoration of service (05:30 – 06:00)

3. Incident Resolution

👉 *The actions taken to restore services and mitigate impact.*

GTM was corrected and 100% of customer traffic was moved to Hall 2 so that the deployment of WCS Release 198 could start on Hall 2

4. Root Cause

👉 *Refers to the fundamental reason (s) why an incident occurred including problem description, analysis techniques, contributing factors, evidence with concise root cause statements*

The root cause was human error whilst making configuration changes for the GTM and while reviewing the changes that had been made.

1. There are seven property changes in the Akamai Configuration for Global Traffic Management (GTM) that need to be made in order to move the customer traffic between halls.

This configuration has to be changed manually via Akamai GUI. However, instead of all 7 properties being modified correctly, only the first property was correctly changed. This resulted in only the first property routing traffic to Hall 1 (i.e. the hall which should receive all customer traffic whilst the deployment took place on hall 2).

2. Although a peer view of the configuration changes took place, instead of validating all 7 properties, only the first property was checked. The reviewer saw that the 1st property was pointing to the correct hall and did not notice that the remaining 6 were pointing to the wrong hall.

5. Findings

👉 *Refer to the key insights and conclusions drawn from the incident analysis - what went well/what went wrong highlighting gaps, failures, or inefficiencies.*

The incident itself was quickly resolved. However, the main issue is that although there are documented processes in place to check and verify configuration changes made during a deployment, these checks were not strictly adhered to.

5.1. Observability –

👉 *Evaluating monitoring, logging, and alerting effectiveness in detecting the incident.*

Alerts were received by various teams (WCS, Identity, Price)

5.2. Technical Challenges -

👉 *Identifying system, infrastructure, or architectural issues that complicated resolution.*

5.3. MIM involvement & impact assessment -

👉 *Timely engagement of MIM and analysing the impact on business operations based on top 10 business processes*

This area is managed by the Service Leads, not the MIM team. The issue was reported in a Teams channel and the on-call Service Lead was called out

5.4. Incident handling/stakeholder management-

👉 *Reviewing how teams collaborated to drive the incident to resolution and engaged with key stakeholders.*

Teams quickly identified that the problem was caused by the on-going WCS release. The WCS team were already aware of the issue when they joined the bridge.
The standard incident comms were used to communicate with key stakeholders.

5.5. Business Communication –

👉 *Clear, concise & timely communication to stakeholders, focusing on business impact and recovery actions.*

Business communication was clear, concise and timely.

5.6. Supplier & Product team Engagement –

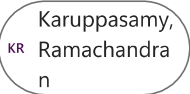
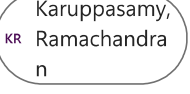
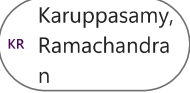
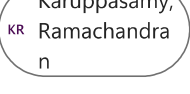
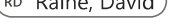
👉 *Assessing the responsiveness and support provided by external vendors and internal product teams.*

N/A

Mitigation Actions:

👉 *Agree actions for each of the findings to prevent recurrence, strengthen system reliability & stability, and improve response efficiency for future incidents*

	≡ Action	👤 Owner	📅 Due date	≡ Remarks/ Updates	🏷️ Action Category
1	Rigor In Operations - Team Refresh session to be done (Not to be overconfident & complacent)	 Sumitro Chakravarty	Tue, Apr...	.	Problem Avoidance
2	Multiple tasks not getting clubbed together by the same person performing the change - GTM Change, Deployment, DB Checks.	 Sumitro Chakravarty	Tue, Apr...		Problem Avoidance

3	Carry out the GTM at around 10:30 PM UK time the previous night so that the person doing the release can check that traffic is 100% on the correct hall. To be added to the deployment plan Note: Ensure that a single hall can handle the traffic before carrying out the GTM	 Karuppasamy, Ramachandra n	Tue, Apr...		Problem Avoidance
4	DBA will check the traffic in the database to ensure that no hit is coming from the hall/origin where the deployment will happen. Add this step to the deployment plan	 Gaikwad, Vivekananda  Karuppasamy, Ramachandra n	Tue, Apr...		Problem Avoidance
5	Review whether an alert can be triggered if the GTM has not worked as expected	 Karuppasamy, Ramachandra n	Tue, Apr...		Problem Avoidance
6	Review whether it is possible to simplify the GTM (i.e. review number of properties, currently 7, can any of them be combined)	 Pandi, Thirumal	Tue, Apr...		Problem Avoidance
7	Check absolute numbers for GTM (visual confirmation). Create New Relic widget to show %age of traffic on each hall	 Karuppasamy, Ramachandra n	Tue, Apr...		
8	Review with David Raine whether or not the GTM process can be automated	 Sumitro Chakravarty  RD Raine, David	Tue, Apr...		Solution & Recovery

Please refer to  Major_ Significant and Key Incident Tracker 16th O... for progress updates on Mitigations.

Technical Details and Useful Documents: